

NR-19 Internal Allocation Rules and Tolerances

Internal rules for May 2026 source reconciliation. Apply only to supplied NR-19 records.

Calculation and Tolerance Rules

rule_id	rule_name	formula_or_limit	unit	status_use
AR-001	Oil-rate comparison	absolute difference <= 10	bopd	SUPPORTED if within limit; CONFLICT if exceeded
AR-002	Water-rate comparison	absolute difference <= 8	bwpd	SUPPORTED if within limit; CONFLICT if exceeded
AR-003	Gross-gas comparison	absolute difference <= 50	Mscfd	SUPPORTED if within limit; CONFLICT if exceeded
AR-004	Gas-lift deduction comparison	absolute difference <= 5	Mscfd	SUPPORTED if within limit; CONFLICT if exceeded
AR-005	Net produced gas	$gross_gas_mscfd - gas_lift_mscfd$	Mscfd	CALCULATED where both inputs are present
AR-006	Water cut	$water_bwpd / (oil_bopd + water_bwpd) \times 100$	percent	CALCULATED where both rates are present
AR-007	Gas-oil ratio	$net_produced_gas_mscfd \times 1000 / oil_bopd$	scf/STB	CALCULATED where net gas and oil are present
AR-008	Liquid rate	$oil_bopd + water_bwpd$	bpd	CALCULATED where both rates are present
AR-009	Productivity index	$liquid_rate_bpd / (datum_pressure_psia - flowing_bhp_psia)$	bpd/psi	CALCULATED only where drawdown is positive and both pressure inputs are present
AR-010	Datum pressure correction	$gauge_pressure_psia + pressure_gradient_psi_ft \times (target_datum_tvdss_ft - gauge_tvdss_ft)$	psia	CALCULATED where all datum inputs are present
AR-011	Datum-pressure tolerance	absolute difference <= 5	psi	CONFLICT if supplied reported datum pressure differs by more than limit
AR-012	Pressure/test date link	absolute date difference <= 2	days	CONFLICT if pressure survey and test date differ by more than limit
AR-013	Voidage	$oil_bbl \times oil_fvf_rb_stb + water_bbl + net_gas_mscf \times gas_reservoir_factor_rb_mscf$	rb	CALCULATED where all factors are explicit
AR-014	Voidage replacement ratio	$injection_volume_bbl / voidage_reservoir_bbl$	fraction	CONFLICT if outside target range
AR-015	Monthly field oil total	absolute difference <= 25	bbl	CONFLICT if field total differs from well-level total by more than limit
AR-016	Monthly field gas total	absolute difference <= 100	Mscf	CONFLICT if field total differs from well-level total by more than limit
AR-017	Monthly injection total	absolute difference <= 100	bbl	CONFLICT if field injection total differs from injection ledger by more than limit

Status Definitions

status_label	definition
SUPPORTED	Value is directly supported by supplied source records and no conflicting supplied value is found.
CONFLICT	Two or more supplied files give different values beyond the applicable tolerance for the same well, date or month, reservoir zone, measurement type, and unit.
MISSING	A required value is blank, absent, or unsupported by supplied records.
CALCULATED	Value is calculated directly from explicit source inputs using a listed rule.
TEXT_ONLY	Record is descriptive and has no numeric quantity.
DUPLICATE	Same source repeats the same well, date or month, measurement type, unit, and value.

Restricted Actions

record_id	restricted_action
AR-PROH-001	Do not create reserve estimates or production forecasts.
AR-PROH-002	Do not recommend workovers, stimulation, gas-lift redesign, or operating changes.
AR-PROH-003	Do not certify allocation, regulatory reporting, commercial reporting, or safety status.
AR-PROH-004	Do not infer missing pressures, test durations, choke sizes, datum values, PVT factors, or gas-lift volumes.